TWO GYSIN PULLBACKS ALONG EFFECTIVE CARTIER DIVISORS COMMUTE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: a reduction to one integral closed subspace, a normalised local computation of the two composites as order weighted sums, and the key formula together with a vanishing check on the boundaries

Editorial note. The label lemma-gysin-commutes-gysin names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The Gysin homomorphism attached to a triple of an invertible sheaf, a meromorphic section and its divisor of zeros, and the induced triples obtained by restricting to a divisor and on the intersection
- The key formula comparing the two ways of writing the difference of two divisor constructions as a boundary, and the local order bookkeeping in henselized local rings at the generic points of the prime divisors
- Rational equivalence for locally finite sums of cycles, which upgrades an equality of cycle representatives produced this way to an equality of rational equivalence classes

Source reference. Original reference 02T8: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 3448–3448. Prerequisite pointer invoked at proof line(s) 3793; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0AYC: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 3085–3085. Prerequisite pointer invoked at proof line(s) 3852; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 02RZ: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 1991–1991. Prerequisite pointer invoked at proof line(s) 3811; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:spaces-chow.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original section 076Y: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 2994–2994. Prerequisite pointer invoked at proof line(s) 3830; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 02QL: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 264–264. Prerequisite pointer invoked at proof line(s) 3790; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In Situation 02QL let X/B be good. Let $(\mathcal{L}, s, i : D \to X)$ and $(\mathcal{L}', s', i' : D' \to X)$ be two triples as in Definition 02T8. Then the diagram*

$$\begin{array}{ccc} \mathrm{CH}_k(X) & \xrightarrow{i^*} & \mathrm{CH}_{k-1}(D) \\ {\scriptstyle (i')^*}\downarrow & & \downarrow \\ \mathrm{CH}_{k-1}(D') & \longrightarrow & \mathrm{CH}_{k-2}(D \cap D') \end{array}$$

commutes where each of the maps is a Gysin map.

Proof. Denote $j : D \cap D' \to D$ and $j' : D \cap D' \to D'$ the closed immersions corresponding to $(\mathcal{L}|_{D'}, s|_{D'}$ and $(\mathcal{L}'_D, s|_D)$. We have to show that $(j')^* i^* \alpha = j^*(i')^*\alpha$ for all $\alpha \in \mathrm{CH}_k(X)$. Let $W \subset X$ be an integral closed subscheme of dimension k. We will prove the equality in case $\alpha = [W]$. The general case will then follow from the observation in Remark 02RZ (and the specific shape of our rational equivalence produced below). We will deduce the equality for $\alpha = [W]$ from the key formula.

We let σ be a nonzero meromorphic section of $\mathcal{L}|_W$ which we require to be equal to $s|_W$ if $W \not\subset D$. We let σ' be a nonzero meromorphic section of $\mathcal{L}'|_W$ which we require to be equal to $s'|_W$ if $W \not\subset D'$. Write

$$\mathrm{div}_{\mathcal{L}|_W}(\sigma) = \sum \mathrm{ord}_{Z_i, \mathcal{L}|_W}(\sigma)[Z_i] = \sum n_i [Z_i]$$

and similarly

$$\mathrm{div}_{\mathcal{L}'|_W}(\sigma') = \sum \mathrm{ord}_{Z_i, \mathcal{L}'|_W}(\sigma')[Z_i] = \sum n'_i [Z_i]$$

as in the discussion in Section 076Y. Then we see that $Z_i \subset D$ if $n_i \neq 0$ and $Z'_i \subset D'$ if $n'_i \neq 0$. For each i, let $\xi_i \in |Z_i|$ be the generic point. As in Section 076Y we choose for each i an element $\sigma_i \in \mathcal{L}_{\xi_i}$, resp. $\sigma'_i \in \mathcal{L}'_{\xi_i}$ which generates over $B_i = \mathcal{O}^h_{W,\xi_i}$ and which is equal to the image of s, resp. s' if $Z_i \not\subset D$, resp. $Z_i \not\subset D'$. Write $\sigma = f_i \sigma_i$ and $\sigma' = f'_i \sigma'_i$ so that $n_i = \mathrm{ord}_{B_i}(f_i)$ and $n'_i = \mathrm{ord}_{B_i}(f'_i)$. From our definitions it follows that

$$(j')^* i^* [W] = \sum \mathrm{ord}_{B_i}(f_i) \mathrm{div}_{\mathcal{L}'|_{Z_i}}(\sigma'_i|_{Z_i})$$

as cycles and

$$j^*(i')^*[W] = \sum \mathrm{ord}_{B_i}(f'_i) \mathrm{div}_{\mathcal{L}|_{Z_i}}(\sigma_i|_{Z_i})$$

The key formula (Lemma 0AYC) now gives the equality

$$\sum \Big(\mathrm{ord}_{B_i}(f_i) \mathrm{div}_{\mathcal{L}'|_{Z_i}}(\sigma'_i|_{Z_i}) - \mathrm{ord}_{B_i}(f'_i) \mathrm{div}_{\mathcal{L}|_{Z_i}}(\sigma_i|_{Z_i}) \Big) = \sum \mathrm{div}_{Z_i}(\partial_{B_i}(f_i, f'_i))$$

of cycles. Note that $\mathrm{div}_{Z_i}(\partial_{B_i}(f_i, f'_i)) = 0$ if $Z_i \not\subset D \cap D'$ because in this case either $f_i = 1$ or $f'_i = 1$. Thus we get a rational equivalence between our specific cycles representing $(j')^* i^*[W]$ and $j^*(i')^*[W]$ on $D \cap D' \cap W$. $\square$